

Capstone Project by CA. Rajat Bhatia

AICA - Level 2 - Batch 85

Summary Document

Software Summary & Functional Overview

1. Executive Summary

The University MOU, Agreement & Contract Centralized Repository is a full-stack institutional application designed to provide a centralized, searchable and secure repository for Memoranda of Understanding (MOUs), agreements, contracts and other institutional partnership documents.

The system combines document storage, automated information extraction, agreement lifecycle tracking, clause completeness guidance, role-based access, reporting and audit trails in one practical platform.

2. Current Implementation Status

The application has been developed as a working software system and tested successfully. The current implementation includes the major functional modules described in this document.

Component	Current Access
Frontend	http://localhost:5173
Backend API	http://localhost:5000

3. User Roles & Access

System Administrator

- Campus-wide visibility of agreements and repository records.
- Create, edit and manage academic departments.
- Create and manage staff user accounts and their department assignments.
- View institutional activity/audit logs.
- Access reporting and CSV export functions.
- Access system-level settings and administration functions.

Department User

- Access is restricted to the user's assigned department.
- Upload and maintain agreements belonging to the assigned department.
- Search, filter, view, edit and download permitted agreement records.
- Cannot access records belonging to other departments.

4. Core Software Functions

4.1 Agreement Repository

- Central repository for MOUs, Agreements, Contracts, MoAs, Partnerships and other institutional documents.
- Stores key metadata including reference number, title, agreement type, department, parties, purpose, dates, status, financial information, summary and remarks.
- Original uploaded documents are retained for later viewing and download.

4.2 Automated Document Reading

- Supports PDF, DOCX and TXT document uploads.
- Documents can be uploaded using a file upload or drag-and-drop workflow.
- The system reads the uploaded document and pre-fills relevant agreement information.
- Users can review, correct or manually enter information before saving the record.

4.3 Information Extracted from Documents

- First Party
- Second Party
- Agreement Type
- Purpose
- Signing Date
- Start Date
- Expiry Date
- Financial Implication
- Financial Amount, where available
- Important Remarks
- Key Clauses

- Summary of the agreement

4.4 Clause Suggestions & Legal Analysis

- The system checks agreements against a 10-point clause checklist.
- Confidentiality
- Intellectual Property (IP)
- Data Protection
- Termination
- Renewal
- Dispute Resolution
- Governing Law
- Liability / Indemnity
- Force Majeure
- Payment / Financial Terms
- Provides guidance notes, a completeness score and suggestions for potentially missing provisions.
- The feature is intended as general informational guidance and does not replace review by the University's authorized legal authority.

4.5 Agreement Lifecycle & Expiry Tracking

- Active
- Expiring Soon — within 90 days
- Expired
- No Expiry Date — perpetual/undated
- Dedicated Expiring Soon view with countdown categories for less than 30, 31–60 and 61–90 days.

4.6 Agreement Details

Each agreement has a structured detail view with seven tabs:

- Overview
- Parties
- Financial Information
- Important Clauses
- Document
- Suggestions

- History
- Agreement information can be edited through the interface.
- Original documents can be downloaded.
- Records can be deleted subject to the user's permissions.

4.7 Analytics, Reports & Export

- Agreements by department
- Agreements by classification/type
- Status distribution
- Financial vs. non-financial agreements
- Execution timeline by year
- Institution-wide CSV export

4.8 Administration & Audit Trail

- CRUD management for academic departments.
- CRUD management for staff users.
- Searchable and filterable activity log.
- Audit events include uploads, updates, downloads, deletions and logins.

5. Main User Workflow

1. Login
2. Open Dashboard
3. Select Add Agreement
4. Upload the original document
5. System reads and extracts relevant information
6. Review and edit the extracted information
7. Save the agreement
8. Record is stored in the database
9. Search and filter the repository
10. Open the agreement details
11. Review clauses, financial information and expiry status
12. Download the original document when required

6. High-Level System Structure

- Frontend: Web-based user interface for login, dashboard, repository, forms, reports and administration.
- Backend: API layer responsible for authentication, authorization, agreement management, document processing and audit functions.
- Database: Stores users, departments, agreement metadata, extracted information, lifecycle status and activity history.
- Document Storage: Maintains uploaded source documents for retrieval and download.
- Document Intelligence: Reads uploaded documents and assists with extracting agreement information and identifying potentially missing standard clauses.

7. Demo / Test Accounts

Account	Role	Department	Purpose
admin@university.edu	System Admin	All Departments	Full institutional access
cs.user@university.edu	Department User	Computer Science & Engineering	Department-scoped testing
law.user@university.edu	Department User	School of Law & Legal Studies	Department-scoped testing
bio.user@university.edu	Department User	Biotechnology	Department-scoped testing
bus.user@university.edu	Department User	Business School	Department-scoped testing

A Switch User control is available in the top navigation to test role and department-scoping behavior.

Security note: Demo credentials should be changed, disabled or removed before production deployment.

8. Expected Institutional Benefits

- Centralized and organized repository for institutional agreements.
- Reduced dependency on scattered departmental files and manual registers.
- Faster retrieval of agreements and important agreement information.
- Better visibility of approaching expiries and renewals.
- Improved awareness of financial commitments and implications.
- Consistent identification of commonly expected agreement clauses.
- Clear separation of administrative and departmental access.

- Improved accountability through audit history and activity tracking.
- Useful institutional reporting for management and administration.

9. Scope & Practical Considerations

- The repository is designed to remain useful even when automated document-reading features are unavailable; extracted and verified information is stored as normal repository data.
- Before production use, representative real university documents should be tested to verify extraction accuracy across different PDF/DOCX formats and document layouts.
- User permissions and department isolation should be tested with actual institutional roles.
- Production deployment should use secure database and document storage, appropriate authentication, backups and access controls.
- The legal clause-suggestion functionality should be treated as decision support only and reviewed by the University's authorized legal/competent authority.

10. Conclusion

The University MOU, Agreement & Contract Centralized Repository provides a practical institutional platform for storing, organizing, reviewing and monitoring agreements. Its combination of role-based access, automated document reading, structured metadata, expiry monitoring, clause guidance, reporting and audit history creates a foundation for more systematic agreement administration across the University.